# FLUID FLOW PHENOMENA

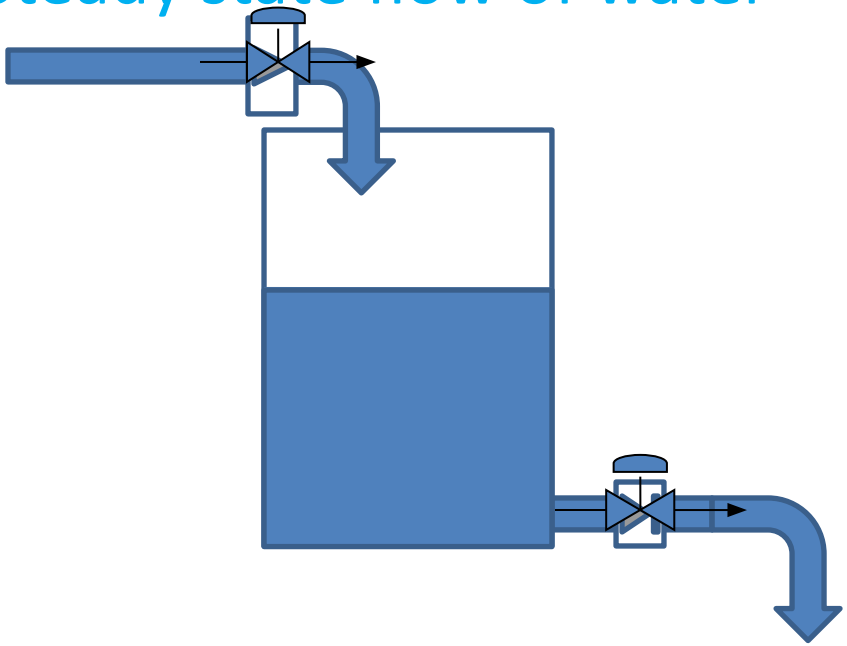
**Compressible and Incompressible Fluids:** If the density is little affected by moderate changes in temperature and pressure, the fluid is said to be *incompressible*, and if the density is sensitive to changes in these variables, the fluid is said to be *compressible*. Liquids are considered to be incompressible and gases compressible.

**Types of Flow**

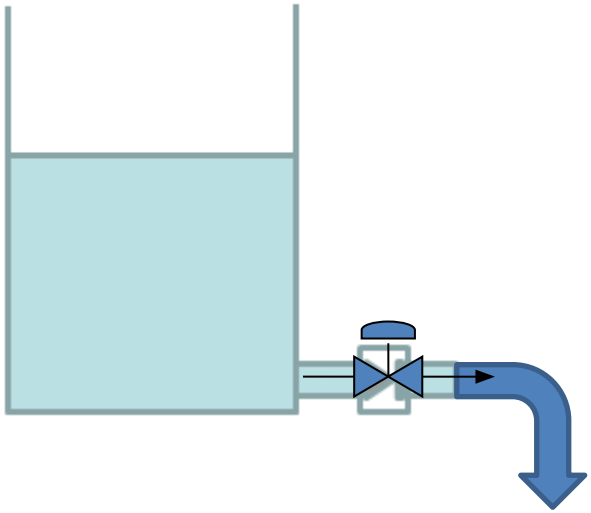
**Uniform Flow**: Flow is described as uniform if the velocity at a given instant is same in magnitude and direction at every point in the fluid..If at given instant, the velocity changes from point to point, the flow is described as non-uniform.

**Steady flow**: A steady flow is one in which the velocity, pressure and cross section of the stream may vary from point to point but do not change with time .If at a given point conditions do change with time, the flow is described as unsteady

Steady state flow of water



Un Steady state flow of water



**In Un steady-state (Transient) process** variables changes with time. Every time we take a snapshot, the variables have different values than in the first snapshot.

# Laminar and Turbulent flow

https://www.youtube.com/watch?v=pae5WrmDzUU

In laminar flows the fluid particles may be considered to be moving in layers or laminae. Laminar flow occurs at low velocities and the flow appears to be smooth without any mixing on a macroscopic scale between adjacent layers. A typical example of laminar flow is the flow of honey or thick syrup from a bottle. Turbulent flows are characterized by a mixing action throughout the flow field caused by eddies in the flow. Thus in turbulent flows the mixing due to molecular motion is superimposed by mixing due to eddies. Though the mixing due to eddies may appear to be small they have important consequences.

Reynolds number is used as a criterion for characterizing the flow as laminar or turbulent. For flow through pipes and ducts, the flow is generally considered to be laminar if the Reynolds number is less than 2100, it is treated as turbulent of Reynolds number exceeds 2100.

**Ideal Fluid and Ideal Gas:** Ideal fluid is frictionless and incompressible but the perfect gas has viscosity and can therefore develop shear stresses and it is compressible according to ideal gas equation of state. Real gases below critical pressure and above the critical temperature tend to obey the perfect gas law.

**Viscosity:** Viscosity is the measure of resistance to flow. For example molasses and tar are highly viscous liquids; water and air have very small viscosities. The viscosity of a gas increases with temperature, but the viscosity of a liquid decrease with temperature.

The resistance of a fluid to shear stress depends upon its cohesion and upon its rate of transfer of molecular momentum. A liquid, with molecules much more closely spaced than a gas, has cohesive force much larger than a gas. Cohesion appears to be the predominant cause of viscosity in liquids; and since cohesion decreases with temperature, the viscosity also decreases. A gas, on the other hand, has very small cohesion forces. Most of its resistance to shear stress is the result of the transfer of molecular momentum and since molecular momentum increases with temperature, the viscosity of a gas also increases with temperature.

For ordinary pressures viscosity is independent of pressure and depends upon temperature only. For very great pressures, gases and most liquids have shown variations of viscosity with pressure.

The SI unit of viscosity is ($N.s/m^2$) or (kg/m.s). The unit of viscosity in CGS system is (g/cm.s) also called poise.
The viscosities (μ) of liquids generally vary approximately with absolute temperature T according to:

$$\ln \mu = a - b \ln T \qquad N_e D V \rho / \mu$$

Viscosity of gases increases with increase in temperature.
The viscosity (μ) of many gases is approximated by the formula:
$\mu = \mu_o (T/T_o)^n$, in which T is the absolute temperature, $\mu_o$ is the viscosity at an absolute reference temperature $T_o$, and n is an empirical exponent that best fits the experimental data.

The viscosity of an ideal gas is independent of pressure, but the viscosities of real gases and liquids usually increase with pressure.

Viscosity of liquids is generally two orders of magnitude greater than gases at atmospheric pressure.

For example, at 25°C,

$\mu_{water} = 1cP$

and $\mu_{air} = 1x10^{-2}\ cP$.

**Kinematic viscosity:** Kinematic viscosity is the ratio of viscosity to density $\gamma = \mu/\rho$. Kinematic viscosity of gases varies with pressure because of change in density. The SI unit of kinematic viscosity is $(m^2/s)$ and the CGS unit is $(cm^2/s)$ called Stoke (St).

**Newton's law of viscosity:** It states that the shear stress is directly proportional to velocity gradient or shear rate.

Let two layers of liquid be separated by a distance dy and moving with velocity du let $\tau$ be the shear stress caused due to the relative motion of layer. According to Newton's law

$$\tau = -\mu \frac{du}{dy}$$



The proportionality factor μ is called the viscosity or coefficient of viscosity.

Fluids are classified based on Newton's law of viscosity as

- Newtonian – fluids which obey Newton's law of viscosity
- Non-Newtonian – fluids which does not obey Newton's law of viscosity.

Non-Newtonian fluids can be again classified as follows

- Time independent
  - ⇒ Bingham plastic fluids
  - ⇒ Pseudoplastic fluids
  - ⇒ Dilatant fluids
- Time dependent
  - ⇒ Thixotropic fluids
  - ⇒ Rheopectic fluids

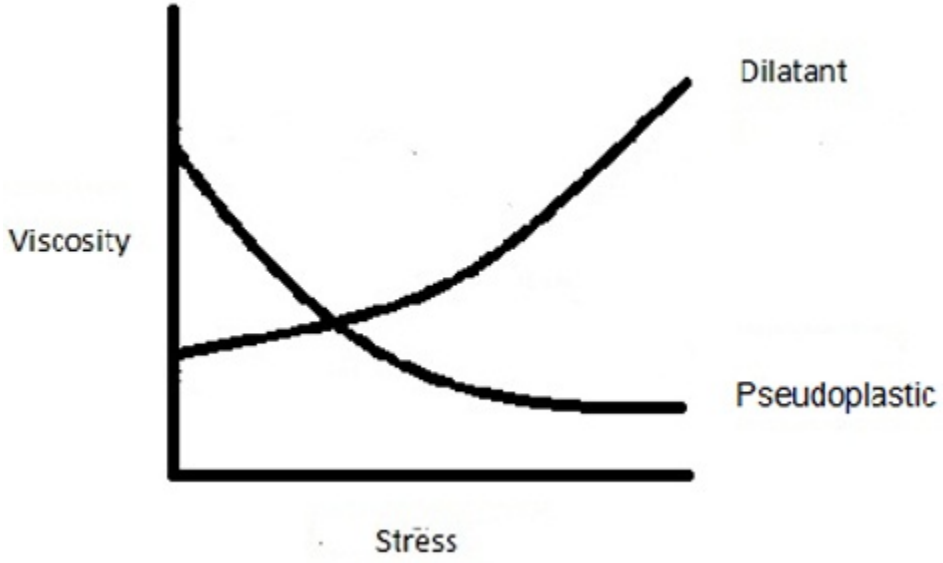
Rheological

Bingham fluids also called Bingham plastics do not flow at all until a threshold shear stress denoted by $\tau_0$ is attained and then flows linearly at shear stress greater than $\tau_0$. It resist a small shear stress indefinitely, but flow easily under large shear stresses. At low stresses the viscosity is infinite and at higher stresses the viscosity is constant.

Examples are bread dough, toothpaste, jellies, and some slurries. Good toothpaste should be a Bingham fluid, so that it can be easily squeezed out of the tube but will not drip off the toothbrush as water or honey would.

**Pseudoplastic fluids** : viscosity decreases with increase in velocity gradient. As shown in figure the curve passes through the origin, and is concave downwards at low shear and is linear at higher shear. Examples are most slurries, mud, polymer solutions, solutions of natural gums, and blood.

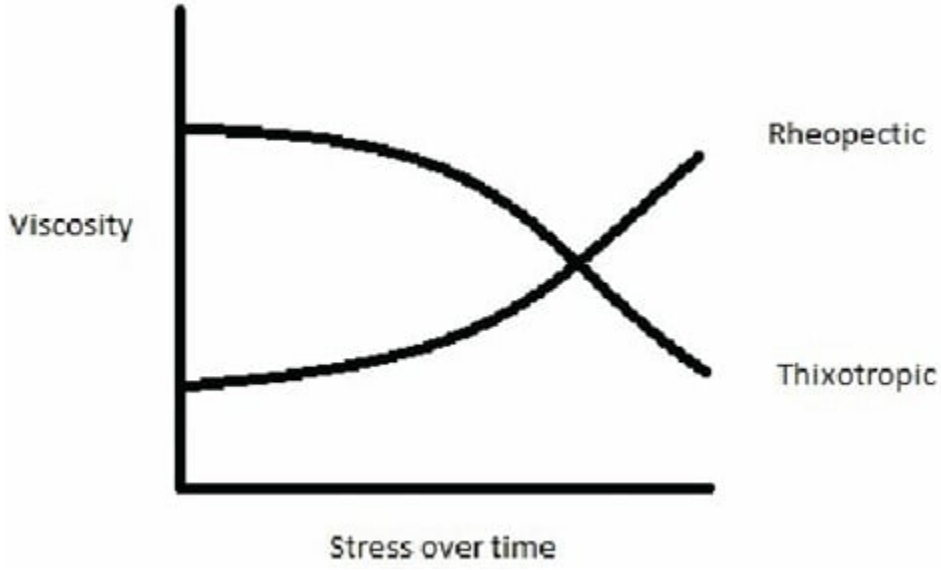
**Dilatant fluids** viscosity increases with increase in velocity gradient. As shown in figure the curve passes through the origin, and is concave upwards at low shear and is linear at higher shear. They are uncommon, but starch suspensions, quick sand etc., behave in this way.



Time dependant flows

If the viscosity variation remain constant with time, in which case the fluid is called time independent. Iet it has been assumed that the curve of τ vs. du/dy in not a function of time. (As explained for Newtonian and Newtonian fluid)

- ✔ The viscosity can decrease with time, at constant shear rate, this type of fluid is called thixotropic.
- ✔ The viscosity can increase with time, at constant shear rate, this type of fluid is called rheopectic.



**Thixotropy** is a time-dependent shear thinning property. Certain gels or fluids that are thick, or viscous, under static conditions will flow (become thin, less viscous) over time when shaken, agitated, sheared or otherwise stressed (time dependent viscosity).

A good paint should be thixotropic, so that in the container it will be very viscous and the pigment will not settle to the bottom, but when it is stirred, it will become less viscous and can be easily brushed onto a surface. In addition, the brushing should temporarily reduce the viscosity, so that the paint will flow sideways and fill in the brush marks (called levelling in the paint industry); then, as it stands, its viscosity should increase, so that it will not form drops and run down the wall and aids easy painting.

**Rheopectic fluids** are a rare class of non-Newtonian fluids. These show a time-dependent increase in viscosity; the longer the fluid undergoes shearing force, the higher its viscosity. Rheopectic fluids, such as some lubricants, thicken or solidify when shaken. Examples of rheopectic fluids include gypsum pastes and printer inks.

A good motor oil should be Rheopectic, so that in the bearing, where the value of du/dy is high, it will offer little frictional resistance and so that at all the gaskets and joints, where the value of du/dy is low, it will be viscous and not leak through.

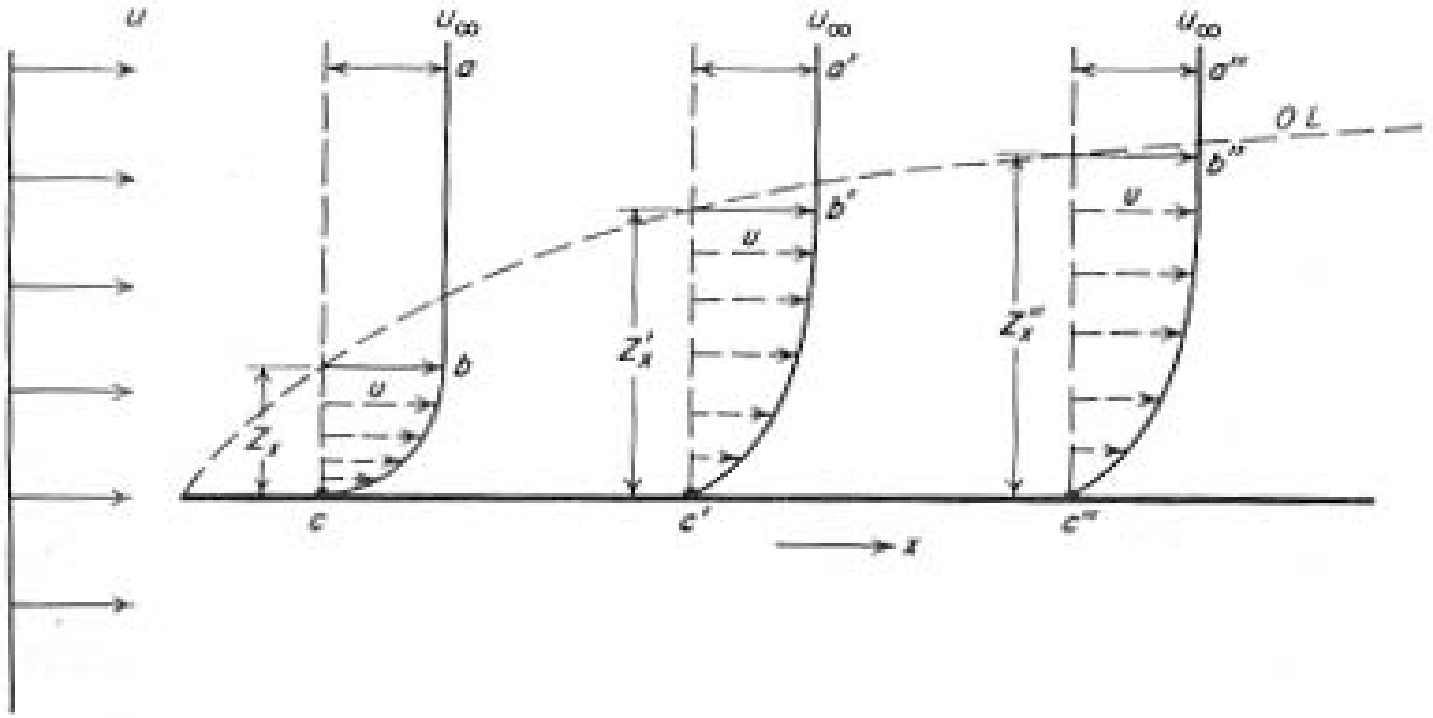


**Free and wall turbulence**

Turbulence can result either from contact of the flowing stream with solid boundaries or from contact between two layers of fluid moving at different velocities. The first kind of turbulence is called wall turbulence and the second kind of turbulence is called free turbulence. Wall turbulence appears when the fluid flows through closed or open channels or past solid shapes immersed in the stream. Free turbulence appears in the flow of a jet into a mass of stagnant fluid (or when a boundary layer separates from a solid wall and flows through the bulk of the fluid).

**Flow in boundary layers:**

A boundary layer is defined as that part of a moving fluid in which the fluid motion is influenced by the presence of a solid boundary. Boundary layer formation considering the flow of fluid parallel with a thin plate as shown in the figure




. Boundary layer formation on a thin plate.

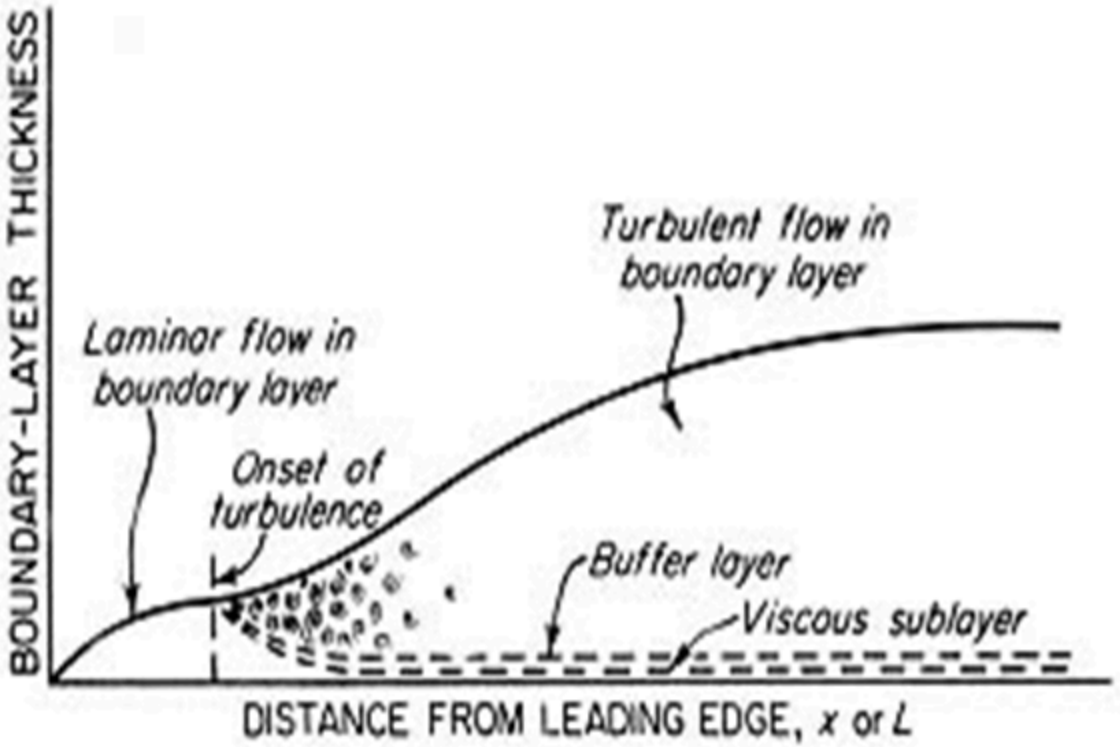
The velocity of the fluid upstream from the leading edge of the plate is uniform across the entire fluid stream. The velocity of the fluid at the interface between the solid and the fluid is zero. The velocity increases with distance from the plate as shown in figure. Each of these curves corresponds to a definite value of x, the distance from the leading edge of the plate. The curves change slope rapidly near the plate. They also show that the local velocity approaches asymptotically the velocity of the bulk of the fluid stream.

The line OL represents an imaginary surface which separates the fluid stream into two parts. One in which the fluid velocity is constant and the other in which the velocity varies from zero at the wall to a velocity substantially equal to that of the undisturbed fluid. This imaginary surface separates the fluid that is directly affected by the plate from that in which the local velocity is constant and equal to the initial velocity of the approach fluid. The zone or layer between the line OL and the plate constitutes the **boundary layer**.

Above OL where there is no influence due to the wall where the velocity is constant is called as **potential flow**

**Laminar and turbulent flow in boundary layer**

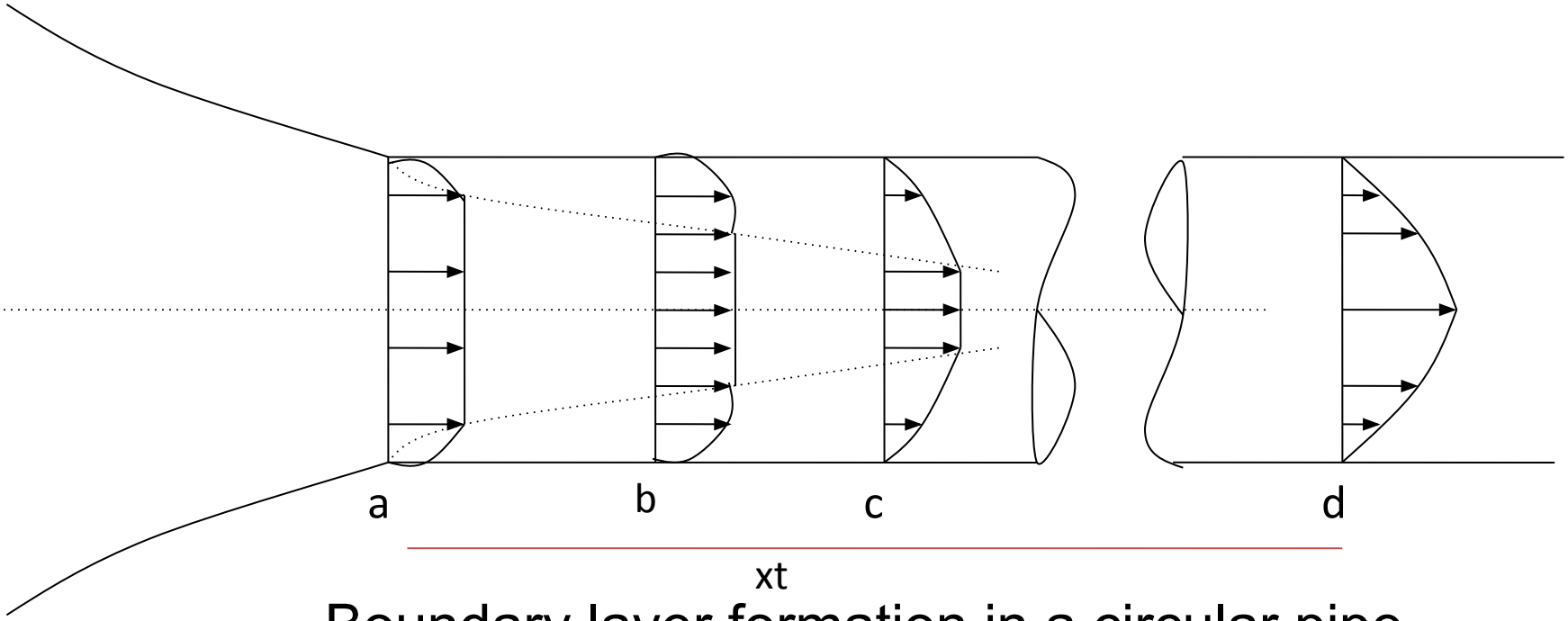
The fluid velocity at the solid interface is zero, and the velocities close to the solid surface are small. Flow in this part of the boundary layer very near the surface therefore is laminar. Further away from the surface the fluid velocities, though less than the velocity of the undisturbed fluid, may be fairly large, and flow in this part of the boundary layer may become turbulent. Between the zone of fully developed turbulence and the region of laminar flow is a transition, or buffer, layer of intermediate character. Thus a turbulent boundary layer is considered to consist of three zone; the viscous sub layer, the buffer layer and the turbulent zone. This is shown in figure




Different region of boundary layers.

The transition length for turbulent flow is less than that for laminar flow unless the Reynolds number is small.

Consider a straight tube having a bell shaped entrance as shown in figure  A boundary layer begins to form at the entrance of the tube, and as the fluid moves through the firs part of the channel, the layer thickens.  During this stage the boundary layer occupies only part of the cross section of the tube, and the total stream consists of a case of fluid flowing in a rod like manner at constant velocity and an annular boundary layer between the wall and core.  In the boundary layer the velocity increases from zero at the wall to the constant velocity existing in the core.  As the stream moves further down the tube, the boundary layer occupies an increasing portion of the cross section.




Boundary layer formation in a circular pipe

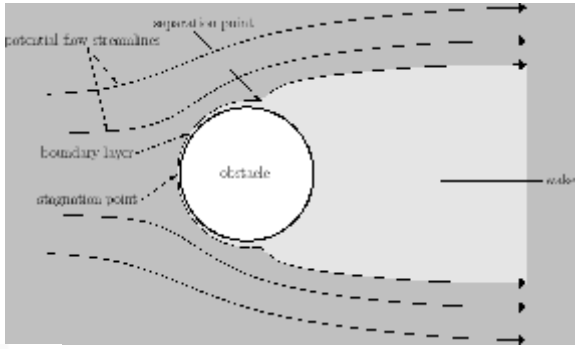
Finally at a point well down stream from the entrance, the boundary layer reaches the centre of the tube and rod like core disappears and the boundary layer occupies the entire cross section of the stream. At this point the velocity distribution in the tube reaches its final form, as shown in figure (position d) and remains unchanged during the remaining length of the tube. Such flow with an unchanging velocity distribution is called Fully developed flow

The length of the entrance region of the tube necessary for the boundary layer to reach the centre of the tube and for fully developed flow to be established is called the transition length. Since the velocity varies not only with length of the tube but with radial distance from the centre of the tube, flow in the entrance region is two dimensional. The approximate length of straight pipe necessary for completion of the final velocity distribution, for laminar flow is

$$\frac{x_t}{D} = 0.05NRe$$

where $x_t$ is transition length and D is diameter of pipe.

# Boundary layer separation and wake formation



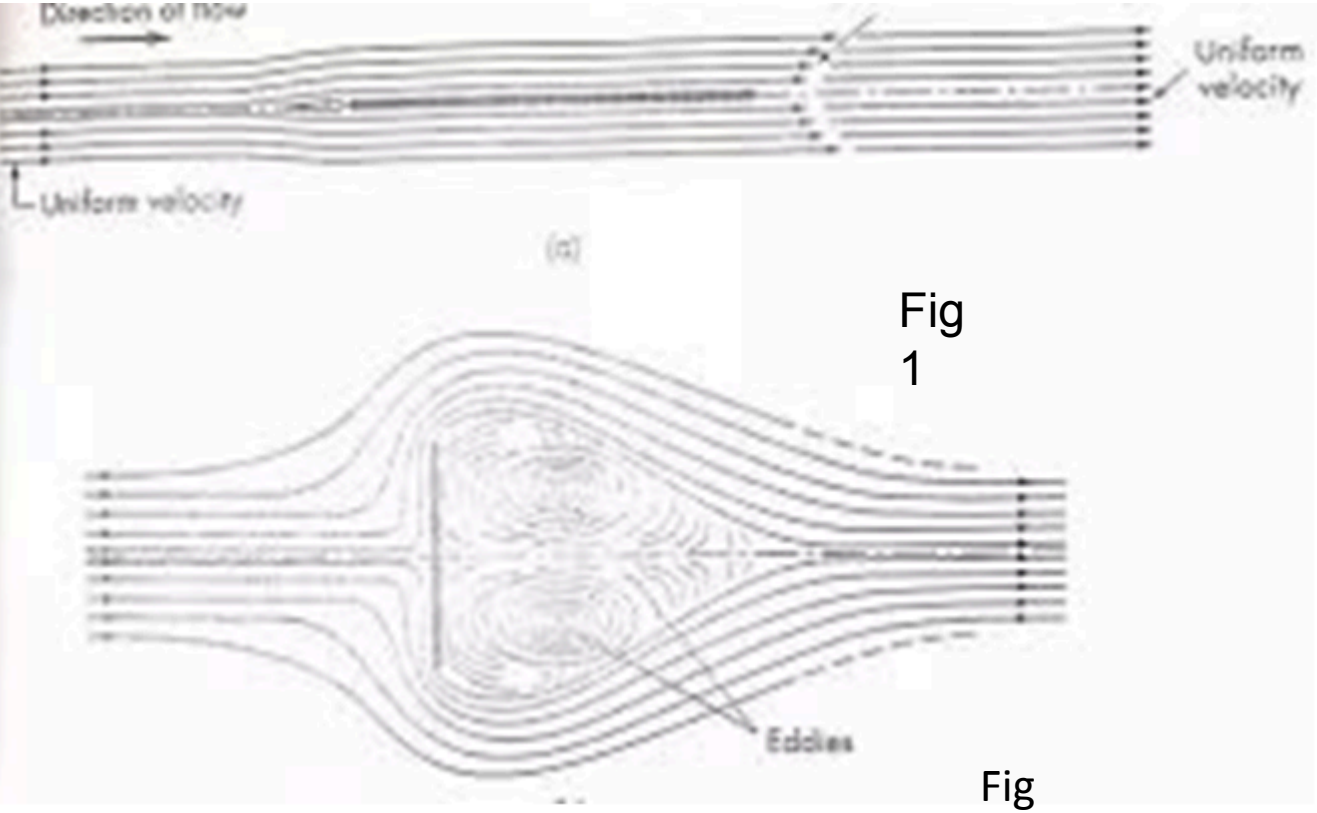




Fig 1

Fig 2

**Boundary layer separation and wake formation**

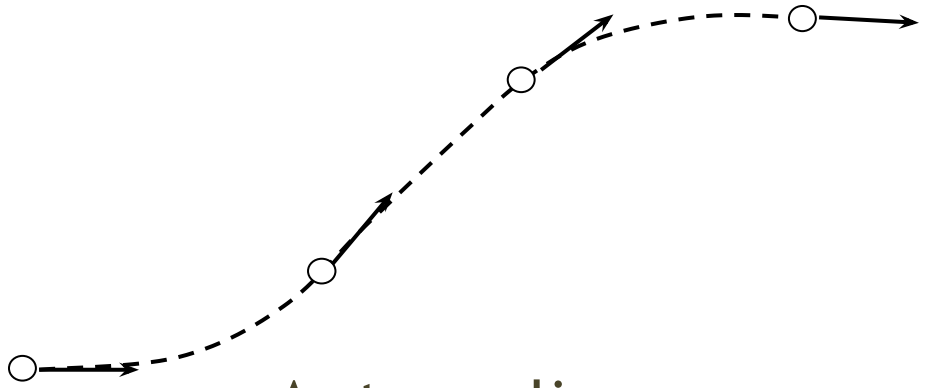
Consider the fluid flow over a flat plate. At the trailing edge of a plate which is parallel to the direction of flow, the boundary layers on the two sides of the plate have grown to a maximum thickness. For a time, after the fluid leave the plate, the layers and velocity gradient persist. Soon, however, the gradients fade out; the boundary layers intermingle and disappear, and the fluid once more move with a uniform velocity. Ref Fig 1

Suppose, now the plate is turned at right angles to the direction of flow over the upstream face, then the fluid reaches the edge of the plate, its momentum prevents it from making the sharp turn around the edge and it separates from the plate and proceeds outward into the bulk of the fluid. Behind the plate is a back water zone of strongly decelerated fluid, in which large eddies, called vortices are formed. This zone is known as the wake.

The eddies in the wake are kept in motion by the shear stresses between the wake and the separated current.  They consume considerable mechanical energy and may lead to a large pressure loss in the fluid. This is shown in Fig 2

# Streamlines

- In analysing fluid flow it is useful to visualise the flow pattern.
- This can be done by drawing lines joining points of equal velocity – (velocity contours).
- These lines are known as *streamlines*.



A streamline

They are velocity vectors drawn in the direction of flow.

## *Explanation of the behaviour streamlines*

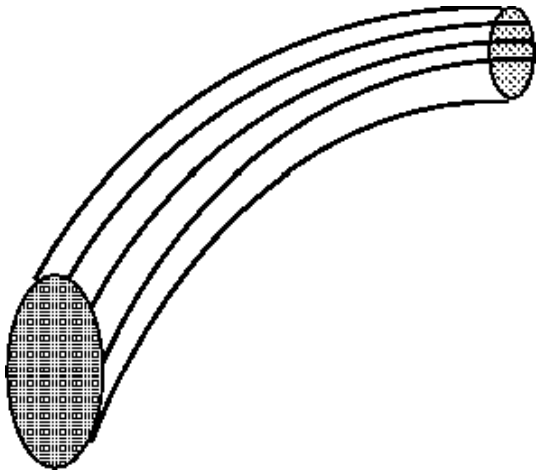
- As fluid is moving in the same direction as the streamlines, fluid can not cross a streamline.
- Streamlines can not cross each other. If they were to cross this would indicate two different velocities at the same point. This is not physically possible.
- It implies that any particles of fluid starting on one streamline will stay on that same streamline throughout the fluid.

# Streamtubes

- A useful technique in fluid flow analysis is to consider only a part of the total fluid in isolation from the rest.
- This can be done by imagining a tubular surface formed by streamlines along which the fluid flows.
- This tubular surface is known as a *streamtube*.
- In a two-dimensional flow, we have a streamtube which is flat (in the plane of the paper).

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- The *"walls"* of a streamtube are made of streamlines.
- As we have seen above, fluid cannot flow across a streamline, so fluid cannot cross a streamtube wall.
- The streamtube can often be viewed as a solid walled pipe.
- A streamtube is not a pipe - it differs in unsteady flow as the walls will move with time. And it differs because the *"wall"* is moving with the fluid.

# Mass flow rate

- $\text{mass flow rate} = m = \dfrac{\text{mass of fluid}}{\text{time taken to collect the fluid}}$

- $\text{time} = \dfrac{\text{mass}}{\text{mass flow rate}}$

# Volume flow rate - Discharge

- $discharge = Q = \frac{volume\ of\ fluid}{time}$

$$= \frac{mass\ of\ fluid}{time} \times \left( density = \frac{mass}{volume} \right)$$

&

$$= \frac{mass\ fluid\ rate}{density} \quad \frac{m}{\rho}$$